

$$1) \theta_i(t) = kt^2.$$

$$L(t) = \frac{d\theta_i}{dt} = 2kt \text{ (linear)}$$

Small error PLL $\theta_e(s) = \frac{s\theta_i(s)}{s + lH(s)}$

(case -1 : $H(s) = 1$)

$$\frac{\theta_e(s)}{\theta_i(s)} = \frac{s}{s+l}, \quad \theta_i(t) = kt^2 \Rightarrow \theta_i(s) = \frac{2k}{s^3}$$

$$\Rightarrow \theta_e(s) = \frac{s}{s+l} \cdot \frac{2k}{s^3} = \frac{2k}{s^2(s+l)}$$

Final value theorem $\lim_{t \rightarrow \infty} \theta_e(t) = \lim_{s \rightarrow 0} s \theta_e(s)$

$$\Rightarrow \lim_{s \rightarrow 0} s \frac{2k}{s^2(s+l)} = \infty$$

Case 2: $H(s) = \frac{s+l_1}{s}$ $\Rightarrow \theta_e(s) = \frac{s^2}{s^2+l_1s+l_2} \theta_i(s)$

$$\Rightarrow \theta_e(s) = \frac{s^2}{s^2+l_1s+l_2} \left(\frac{2k}{s^3} \right)$$

$$\Rightarrow \lim_{t \rightarrow \infty} \theta_e(t) = \lim_{s \rightarrow 0} s \frac{2k}{s(s^2+l_1s+l_2)} = \frac{2k}{l_2} = \text{constant}$$

(case 3): $H(s) = \frac{s^2 + a_1s + b_1}{s^2}$

$$\Rightarrow \theta_e(s) = \left(\frac{s^3}{s^3 + l_1s^2 + l_2s + l_3} \right) \frac{2k}{s^3}$$

$$\Rightarrow \lim_{s \rightarrow 0} s \theta_e(s) = 0$$

$\rightarrow$ zero phase error.

2) $1 \text{ to } 30 \text{ MHz}$. $f_{RF} = 8 \text{ MHz}$.

$$f_c = 10 \text{ MHz}$$

$$f_{RF} = |f_{LO} - f_{RF}| \Rightarrow f_{RF} = f_{LO} - f_{LO}$$

$$1 \leq f_{RF} \leq 30 \Rightarrow 1 \leq f_{LO} - f_{RF} \leq 30$$

$$\Rightarrow 1 \leq f_{LO} - 8 \leq 30 \Rightarrow 9 \leq f_{LO} \leq 38$$

at $f_c = 10 \text{ MHz}$. $f_{image} = 2f_{RF} + f_c = 2(8) + 10$
 $= 16 + 10 = 26 \text{ MHz}$

3) $\phi(t) = 5 \cos(\omega_1 t) + 20 \sin(10000\pi t) + 10 \sin(20000\pi t)$

$$\omega_1 = 2\pi \times 10^5, \quad \beta_1 = 20, \quad \beta_2 = 10$$

$$\Delta \theta = 20 + 10 = 30 \text{ radians}$$

$$\frac{1}{2\pi} \frac{d\phi}{dt} = \omega_c + 20 \cos(10000\pi t) 10000$$

$$+ 10 \cos(2000\pi t) 2000\pi$$

Peak deviation $\rightarrow 20(10000) + 10(2000\pi) = 40000\pi$

$$f_{max} = \frac{40000\pi}{2\pi} = 20000 \text{ Hz}$$

$$f_m = 1000 \text{ Hz} \quad (2000\pi \text{ rad})$$

Carson's rule $\rightarrow B_{FM} = 2B(\beta+1) = 2(20+1)(1000) + 42000 \text{ Hz}$

6) $m(t) \cos(2\pi f_c t + \phi)$

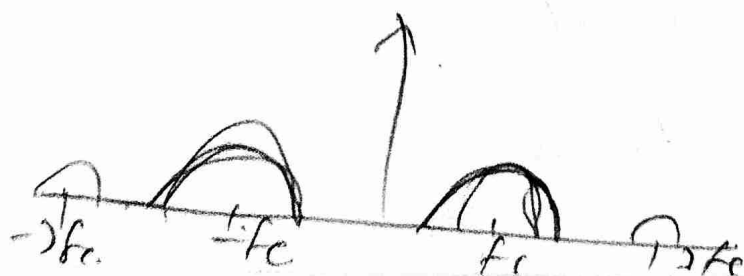
$$\cos^3(2\pi f_c t) = \frac{3}{4} \cos 2\pi f_c t + \frac{1}{4} \cos 6\pi f_c t$$

at mixer output $= m(t) \cos^3 2\pi f_c t$

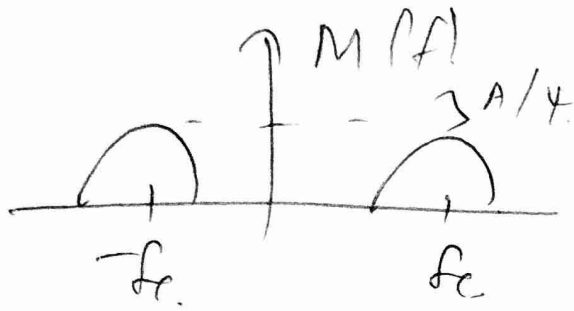
$$= \frac{3}{4} m(t) \cos 2\pi f_c t + \frac{1}{4} m(t) \cos 6\pi f_c t$$

must remove $6\pi f_c t$ term $\rightarrow$ B.P.F at f_c

6)



x)



no overlap b/w $f_c + B$ & $3f_c - B$.

$$3f_c - B > f_c + B \Rightarrow 2f_c > 2B.$$

$$\Rightarrow f_c > B.$$

if outputs $\sin^2(2\pi f_c t) = \frac{3}{4} \sin(2\pi f_c t) - \frac{1}{4} \sin(4\pi f_c t)$

Filter out the term using B.P.F.

if carrier is $\cos^n(2\pi f_c t)$

if n is even $\Rightarrow$ no $\cos 2\pi f_c t$ term.

$\hookrightarrow$ can't do DSB-SC at f_c

if n is odd $\Rightarrow$ always have $\cos 2\pi f_c t$ term

$\hookrightarrow$ DSB-SC at f_c can be done.

$$s) M(f) = \begin{cases} j 2\pi f & |f| < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{for FM} \Rightarrow \phi(t) = K_f \int_{-\infty}^t m(\tau) d\tau$$

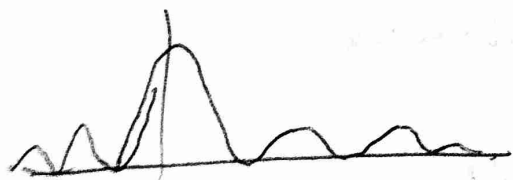
$$K_f = 1 \Rightarrow \phi(t) = \int_{-\infty}^t m(\tau) d\tau$$

$$m(t) = \int_{-\infty}^{\infty} M(f) e^{j 2\pi f t} df$$

$$j 2\pi f \xleftrightarrow{F} \frac{d}{dt} S(t)$$

$$\Rightarrow j 2\pi f \cdot \text{rect}\left(\frac{f}{2}\right) \rightarrow \frac{d}{dt} (2 \text{sinc}(2t))$$

$$\phi(t) = \int m(t) dt = \int \frac{d}{dt} (2 \text{sinc}(2t)) dt = 2 \text{sinc}(2t)$$



$$f_i(t) = f_c + \frac{1}{2\pi} \frac{d}{dt} \phi(t)$$

$$\frac{d\phi}{dt} \propto m(t) \Rightarrow f_i(t) = f_c + \frac{m(t)}{2\pi}$$

$$m\left(\frac{1}{4}\right) = \frac{d}{dt} \left(\frac{1}{2} \sin(2t) \right) = \frac{d}{dt} \left(\frac{\sin 2\pi t}{2\pi t} \right)$$

$$= \frac{1}{2\pi} \cdot \frac{t \cos 2\pi t (2\pi) - \sin 2\pi t}{(t^2)}$$

$$= \frac{1}{2\pi} \left(-\frac{1}{(\sqrt{4})^2} \right) = -\frac{8}{\pi}$$

$$\Delta f = \frac{|m(f)|}{2\pi} = \frac{8/\pi}{2\pi} = \frac{4}{\pi^2}$$

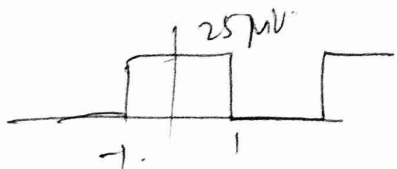
$$B(FM) = 2B + 2\Delta f_{\max} = 2(1) + 2\left(\frac{4}{\pi^2}\right) = 2 + \frac{8}{\pi^2}$$

$$b) \phi(t) = f_c + k_v m(t)$$

$$\frac{1}{2\pi} \frac{d\phi}{dt} = k_v m(t) \Rightarrow m(t) = \frac{1}{2\pi k_v} \frac{d\phi}{dt}$$

triangular pulse $\rightarrow$ square

$$\frac{d\phi}{dt} = \pm 10\pi \quad m(t) = \frac{1}{2\pi \times 2 \times 10^6} = 25\mu V, \quad \frac{d\phi}{dt} = \frac{10\pi}{4\pi \times 10^6}$$



$$\Delta f = \frac{2 \times 10^{16} \text{ Hz} \times 2.5 \times 10^{-6}}{10^{-3}} = 5 \times 10^3$$

$$f_m = 0.5 \text{ Hz}$$

$$B(FM) = 2(B + \Delta f_m) = 2(5 \times 10^3 + 0.5)$$

$$B \approx 10 \text{ kHz}$$

$$7) m(t) \rightarrow B.W = 8 \text{ Hz}, \quad \Delta f = 100 \text{ Hz}, \quad \beta = \frac{\Delta f}{B} = \frac{25}{2}$$

$$B.W \text{ of FM} = 2(\beta + 1)B = 216 \text{ Hz}$$

$$64 \text{ Hz B.P.F} \rightarrow f_c \pm 32 \text{ Hz band}$$

Transmitting signal $\sum J_n(\beta) \cos(2\pi f_c t + n 2\pi f_m t)$

$$\therefore f + n f_m = f_c \pm 32 \rightarrow n = 0, 1, 2, 3, \dots \text{transmits 8}$$

Fraction of power

at B.P.F

$$\approx \frac{|J_0|^2}{2} \approx 2(J_1^2 + J_2^2 + J_3^2 + J_4^2) / 2$$

$$\approx 38.5\%$$

$$8) B = 1 \text{ Hz}, \quad \Delta \beta = 0.1 \quad B.W = 2(0.1 + 1) \times 1 = 2.2 \text{ Hz}$$

multiplication

scales by $\frac{1}{2}$, shifts the signal +1 kHz

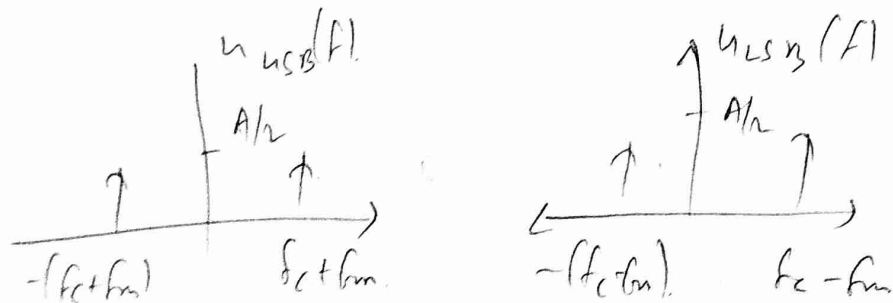
$$\therefore 1/2 \text{ multiplication } y(t) = \frac{1}{2} \cos(2\pi \times 1/2) \times (20)^{1/2}$$

$$9) m(t) = \cos(2\pi f_m t + \phi)$$

$$u_p(t) = A m(t) \cos(2\pi f_c t), \quad f_c > f_m$$

$$u_{USB}(f) = \frac{A}{2} \delta(f - f_c - f_m) + \frac{A}{2} \delta(f + f_c + f_m)$$

$$u_{LSB}(f) = \frac{A}{2} \delta(f - f_c + f_m) + \frac{A}{2} \delta(f + f_c - f_m)$$



$$u_{USB}(t) = \frac{A}{2} \cos(2\pi(f_c + f_m)t + \phi)$$

$$u_{LSB}(t) = \frac{A}{2} \cos(2\pi(f_c - f_m)t + \phi)$$

$$10) x(t) = A_c \cos(2\pi f_c t) \quad m(t) = A_m \cos 2\pi f_m t$$

$$A_c = 10, \quad A_m = 3, \quad f_c = 1 \text{ MHz}, \quad f_m = 10 \text{ kHz}$$

$$x(t) = A_c [1 + \mu \cos(2\pi f_m t)] \cos 2\pi f_c t \quad \mu = \frac{A_m}{A_c} = \frac{3}{10} = 0.3$$

$$x(t) = 10 [1 + 0.3 \cos(2\pi \times 10^4 t)] \cos(2\pi \times 10^6 t)$$

$$P_c = \frac{A_c^2}{2} = \frac{(10)^2}{2} = 50$$

$$P_{total} = P_c \left(1 + \frac{\mu^2}{2}\right) = 50 \left(1 + \frac{0.3^2}{2}\right) = 57.25$$

$$P_{SB} = P_T - P_c = 7.25 \quad \text{each sideband} = \frac{P_{SB}}{2} = 3.625$$

if $\mu = 1.2, \mu > 1 \Rightarrow$ over modulation.
demodulation distortion.

$$11) s(t) = 10 \sin(2\pi f_c t + 20 \sin(2000\pi t))$$

$$f_c = 8 \times 10^6 \quad \beta = 20 \quad f_m = \frac{2000\pi}{2\pi} = 1000 \text{ Hz}$$

$$\Delta f = \beta f_m = 20 \times 1000 = 20 \text{ kHz}$$

$$P = \frac{A^2}{2} = \frac{10^2}{2} = 50$$

$$12) x(t) = A_c \cos(2\pi f_c t), \quad A_c = 8$$

$$m(t) = A_m \cos(2\pi f_m t), \quad A_m = 4$$

$$f_c = 1 \text{ MHz}, \quad f_m = 5 \text{ kHz}, \quad k_f = 10 \text{ kHz}$$

$$x(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$$

$$\Delta f = k_f A_m = 10 \times 10^3 \times 4 = 40 \text{ kHz}, \quad \beta = \frac{\Delta f}{f_m} = \frac{40 \text{ kHz}}{5 \text{ kHz}} = 8$$

$$\Rightarrow x(t) = 8 \cos(2\pi \times 10^6 t + 8 \sin(2\pi \times 5 \times 10^3 t))$$

$$B(\text{FM}) = 2(\beta + 1)B = 2(16 + 1)5 = 90 \text{ kHz}.$$

if A_m doubles, $A_m = 8$.

$$\hookrightarrow \Delta f = 10 \times 10^3 \times 8 = 80 \text{ kHz} \quad / \quad \beta = \frac{80 \text{ kHz}}{5 \text{ kHz}} = 16.$$

$$B = 2(16 + 1)5 = 170 \text{ kHz}.$$

$$13) \mu = \frac{E_{\max} - E_{\min}}{E_{\max} + E_{\min}} = \frac{30 - 10}{30 + 10} = 0.5.$$

$$P_c = \frac{P_c^2}{2} \Rightarrow P_T = P_c \left(1 + \frac{\mu^2}{2}\right) \Rightarrow P_c = \frac{900}{2} = 450.$$

$$P_{\text{total}} = 450 \left(1 + 0.25\right) = 506.25.$$

$$B = 2f_m = \frac{2 \times 1}{0.5 \times 10^{-3}} = 4 \times 10^3 = 4 \text{ kHz}.$$

$$14) m(t) = 2 \sin 2000\pi t - 3 \cos 4000\pi t.$$

$$\mu = 0.7 \quad f_c = 550 \text{ kHz}.$$

$$f_1 = \frac{2000\pi}{2\pi} = 1000, \quad f_2 = \frac{4000\pi}{2\pi} = 2000.$$

$$\eta = \frac{\mu^2/2}{1 + \mu^2/2} = \frac{0.49/2}{1 + 0.49/2} \approx 0.197 \rightarrow \text{power efficiency}$$

$$P_{\text{total}} = 10.$$

$$P_c = \frac{P_T}{1 + \mu^2/2} = \frac{10}{1 + 0.49/2} \approx 8.03.$$

$$PSB = 10 - 8.03 = 1.97.$$

$$\text{each sideband} = 0.985.$$

$$\hookrightarrow \text{side bands} \Rightarrow (580 \pm 1) \text{ kHz}, (580 \pm 2) \text{ kHz}.$$

$$\text{power per sideband} = \frac{0.985}{2} \approx 0.4925.$$

$$15) v_2(t) = a_1 v_1(t) + a_2 v_1^2(t).$$

$$v_1(t) = A_c [1 + k_a m(t)] \cos 2\pi f_c t.$$

$$v_1^2(t) = A_c^2 (1 + k_a m(t))^2 \cos^2 2\pi f_c t.$$

$$\Rightarrow A_c^2 (1 + 2k_a m(t) + k_a^2 m^2(t)) \cos^2 2\pi f_c t.$$

$$\Rightarrow A_c^2 (1 + 2k_a m(t) + k_a^2 m^2(t)) \left(\frac{1 + \cos 4\pi f_c t}{2} \right)$$

$$\Rightarrow v_2(t) = a_1 A_c [1 + k_a m(t)] \cos 2\pi f_c t + a_2 A_c^2 (1 + 2k_a m(t) + k_a^2 m^2(t)) \left(\frac{1 + \cos 4\pi f_c t}{2} \right).$$

passing thr. L.P.F.

$$\Rightarrow V_c(t) = a_c \frac{A_c^2}{2} (1 + 2k_a m(t) + k_a^2 m^2(t))$$

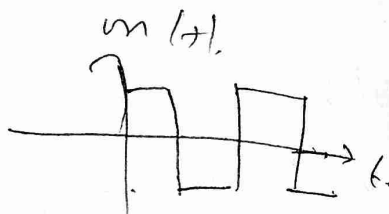
to recover $m(t)$ $k_a m(t) \ll 1$

$m^2(t)$ should be negligible

↳ remove DC term. $\Rightarrow V_{out} \propto m(t)$

$$b) P(t) = \mathcal{U}\left[-\frac{1}{2}, \frac{1}{2}\right](t)$$

$$m(t) = \sum_{-\infty}^{\infty} (-1)^n p(t-n)$$

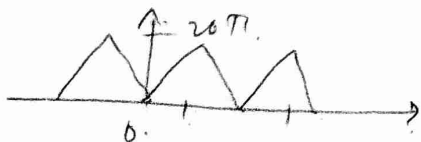


$$\phi(t) = 20 \int_{-\infty}^t m(\tau) d\tau + a$$

$m(t) = \pm 1$ (square wave).

↳ integral $\Rightarrow$ triangular wave.

choose $a \Rightarrow \phi(0) = 0 \Rightarrow a = 0$.



$$\Delta f = \frac{20\pi}{2\pi} = 10$$

$$B = 2(Bf + B) = 2(10 + 2) = 24 \text{ Hz}$$

narrow B.P.F. $\downarrow f_c + \alpha$. $\delta_0 = \frac{1}{2} = 0.5$
($\because \gamma = 2$)

$$(\rightarrow f_c + n(0.5))$$

$$\alpha = 0.5 \Rightarrow n = 1$$

$$\alpha = 0.75 \Rightarrow n = 1.5 \text{ (not integer)}$$

$$\alpha = 1 \Rightarrow n = 2$$

α at 0.75 gives zero power.

$$1) S(t) = A_c \cos(2\pi f_c t + k_p m(t))$$

$$\frac{dS(t)}{dt} = A_c \sin(2\pi f_c t + k_p m(t)) \cdot (2\pi f_c + k_p \frac{dm(t)}{dt})$$

$$A = A_c (2\pi f_c + k_p \frac{dm(t)}{dt})$$

↳ envelope varies proportional to $\frac{dm(t)}{dt}$

PM $\rightarrow$ FM.

envelope detection

$$\Rightarrow 2\pi f_c A_c + A_c k_p \frac{dm(t)}{dt} \text{ remove DC term}$$

Integrate $\rightarrow m(t)$ (message recovered)